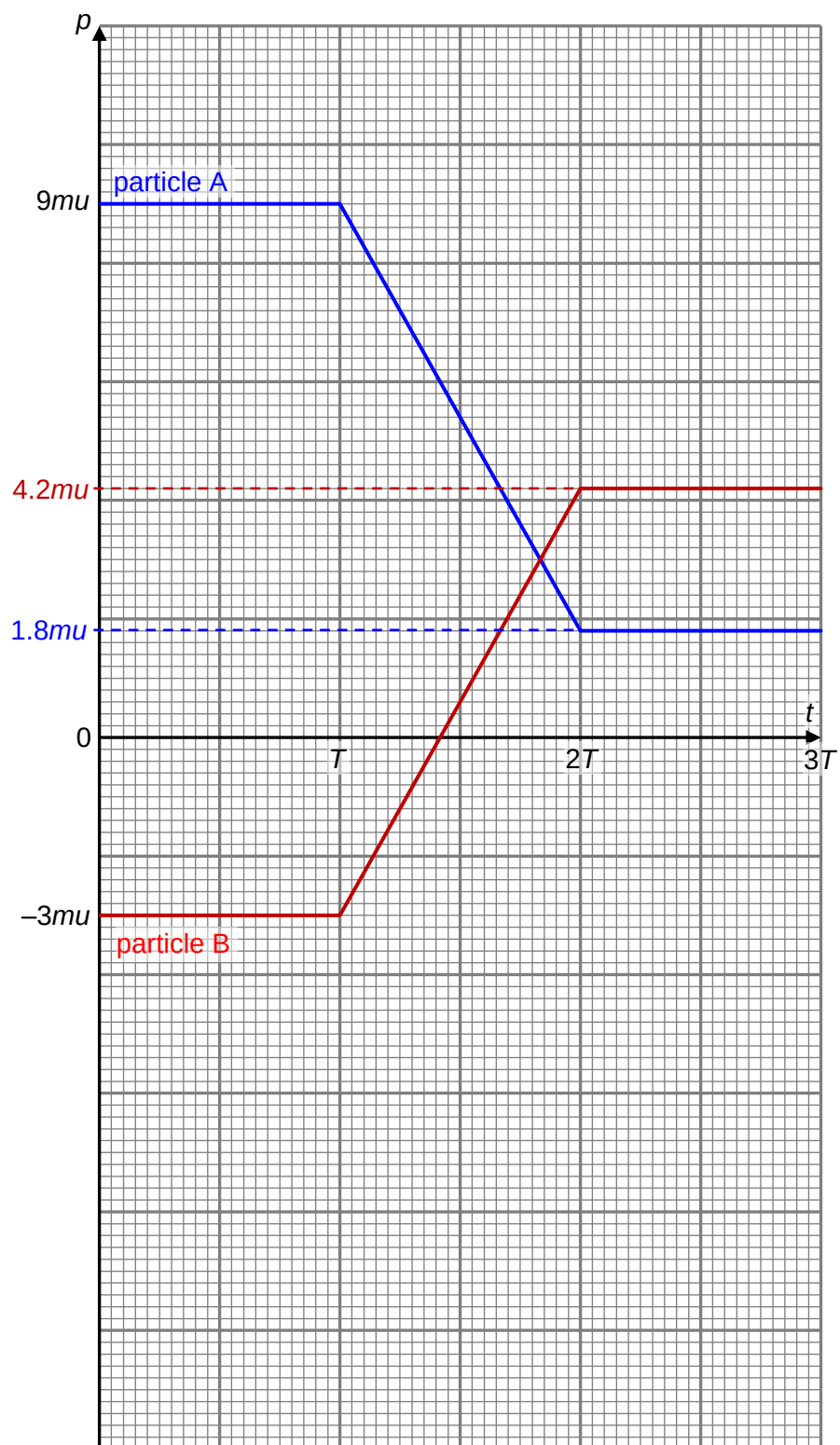


1 (a) (i)



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- (ii) As there are no resultant external forces acting in the horizontal direction on the system of particles A and B, total momentum of the system is conserved in the horizontal direction by the principle of conservation of momentum.

Since the momentum of particle A decreases by $7.2mu$ (36 small squares) after the collision, the momentum of particle B should increase by $7.2mu$ (36 small squares) so that the total momentum remains constant at all times.

OR

Since the momentum before the collision is $6.0mu$, the total momentum of the system remains constant at $6.0mu$ at all times.

(b) Solution 1

$$\text{relative speed of approach} = u_A - u_B = u - (-3u) = 4u$$

$$\text{relative speed of separation} = v_B - v_A = 4.2u - 0.2u = 4u$$

Since the relative speed of approach and the relative speed of separation are equal, the collision is elastic.

Solution 2

$$\text{kinetic energy, } E_k = \frac{1}{2}mv^2 = \frac{1}{2} \frac{(mv)^2}{m} = \frac{p^2}{2m}$$

before collision:

$$E_{k,\text{before}} = \frac{p_{A,\text{before}}^2}{2m_A} + \frac{p_{B,\text{before}}^2}{2m_B} = \frac{(9mu)^2}{2(9m)} + \frac{(-3mu)^2}{2m} = 9mu^2$$

after collision:

$$E_{k,\text{after}} = \frac{p_{A,\text{after}}^2}{2m_A} + \frac{p_{B,\text{after}}^2}{2m_B} = \frac{(1.8mu)^2}{2(9m)} + \frac{(4.2mu)^2}{2m} = 9mu^2$$

Since the kinetic energy of the system before and after the collision remains the same, the collision is elastic.

- (c)** By Newton's second law, resultant force $F_R = \frac{dp}{dt}$, which is given by the gradient of the graph in Fig. 1.1.

During collision, from $t=T$ to $t=2T$, the gradient of the graph for particle A is $-\frac{7.2mu}{T}$. The gradient of the graph for particle B is $\frac{7.2mu}{T}$.

Hence the horizontal resultant force on particle A, which is the force on A by B has magnitude $\frac{7.2mu}{T}$ and the horizontal resultant force on particle B, which is the force on B by A also has the same magnitude of $\frac{7.2mu}{T}$.

Since the gradients have opposite signs, this imply that the two forces are opposite in direction.

This is consistent with Newton's third law as the force on A by B and the force on B by A are equal in magnitude and opposite in direction.

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